Linear Time-Varying Damped Oscillator Verification

Verdict summary

Overall verdict: FAIL: energy_ledger_residual

Check	Observed value	Tolerance / criterion	Verdict
d/dt[A y'] = A y'' + A' y'	0	exactly 0	PASS
Energy identity	0	exactly 0	PASS
A ≡ 1 reduction	0	exactly 0	PASS
A ≡ a reduction	0	exactly 0	PASS
A ≡ 1 limiting max error	3.724298647256319e-12	<= 1e-10	PASS
A ≡ a limiting max error	1.318126163774025e-12	<= 1e-10	PASS
Energy-ledger residual at finest grid	0.00017968732425515527	<= 1e-6	FAIL
eta = 1.0 violates [1/phi, phi]	False	False	PASS

Methods

The equation was written in the state form y'=v and v'=(F-A'v-γv-ω0²y)/A. The manufactured solution y_ex(t)=exp(-0.1t) cos(2t)+0.3 sin(3t) was used to compute the forcing term exactly with gamma = 0.2 and omega0 = 1.5. The custom RK4 and explicit midpoint methods were implemented directly and verified against SciPy DOP853 with rtol=1e-13 and atol=1e-15. The fixed tolerance rules were defined before running: symbolic residual must be exactly zero, limiting-case error <= 1e-10, convergence order within 0.1 of theory, and the energy-ledger residual threshold is 1e-6. The seed for the random energy study is 0.

Symbolic checks

Check	Residual	Verdict
d/dt[A y'] = A y'' + A' y'	0	PASS
Energy identity	0	PASS
A ≡ 1 reduction	0	PASS
A ≡ a reduction	0	PASS

Convergence summary

Case	N	RK4 end error	RK2 end error	RK4 order	RK2 order
A1	1600	4.286e-11	3.860e-05	3.995	2.007
A2	1600	1.064e-11	1.237e-05	4.003	2.003

A_eta	1600	3.757e-11	2.874e-05	3.997	1.995

Coefficient bounds

eta	min A	max A	min A'	A in [1/phi, phi]?
0.000	1.000000	1.000000	0.000000	YES
0.250	0.826630	1.209732	-0.252605	YES
0.481	0.693164	1.442659	-0.499805	YES
1.000	0.466921	2.141688	-1.168215	NO

Energy and limiting-case checks

Quantity	Value	Verdict
Energy residual max	2.851e-03	FAIL
A1 limiting max error	3.724e-12	PASS
A=a limiting max error	1.318e-12	PASS
Underdamped freq.	1.059481005021	PASS

Floquet summary

gamma=0.0: |lambda| = [0.999999999999996, 0.999999999999996]; det error = 8.105e-15
gamma=0.1: |lambda| = [0.7215876121651945, 0.7215876121651945]; det error = 3.020e-14
gamma=0.2: |lambda| = [0.5206886820303004, 0.5206886820303004]; det error = 2.831e-15
gamma=0.39: |lambda| = [0.2678053247348457, 0.2929783503514257]; det error = 2.345e-15
gamma=0.5: |lambda| = [0.19563445476482916, 0.19563445476482916]; det error = 2.616e-15

Discrepancies and notes

The exact symbolic identities hold. The manufactured forcing term was corrected to the exact second derivative of the chosen solution, because the initial draft used a mismatched expression. The energy-ledger accumulation check is the one failed item: the symbolic identity is exact, but the cumulative trapezoidal integral of the power does not bring the numerical energy residual below the fixed tolerance under refinement.

Reproducibility

Platform: Windows 11 / x86_64. Python: 3.13.7; NumPy: 2.4.4; SciPy: 1.17.1; SymPy: 1.14.0; Matplotlib: 3.10.9; ReportLab: 4.4.4. Seed: 0. Command: c:/python313/python.exe oscillator_verification.py. Test command: c:/python313/python.exe -m pytest -q.